



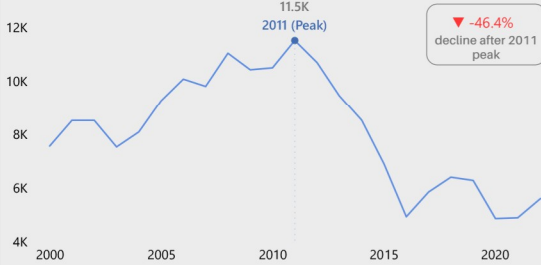
Regional Vulnerability Following Industrial Decline

An exploratory analysis of mining, labor-market, and demographic change in West Virginia, 2011–2024

What Happened to Mining Employment?

West Virginia mining employment peaked in 2011 and declined substantially afterward, reaching a level 46% below the peak by 2024.

Statewide Mining Employment in West Virginia
(Thousands of Jobs)

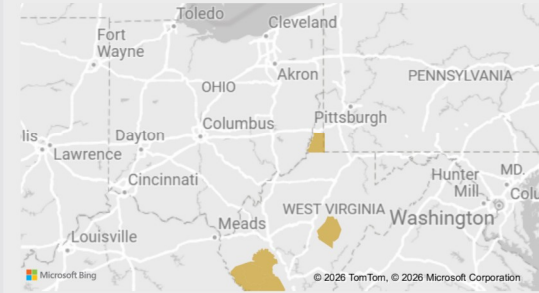


Source: MSHA Mine Employment Data

Which Counties Were Studied?

Seven counties met the criteria for substantial mining employment in 2011.

Coal-Dependent Counties (2011)

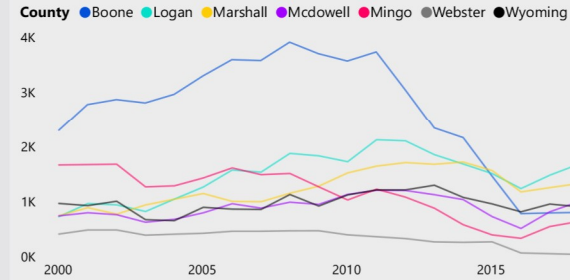


Counties were selected based on $\geq 10\%$ mining employment share and > 200 mining employees in 2011.

Did Mining Employment Change Similarly?

The decline varied substantially across the seven counties.

County Mining Employment in West Virginia
(Thousands of Jobs)



Source: MSHA Mine Employment Data

Why Use 2011?

2011 is the common statewide reference year. Specific county peak years are shown separately because individual counties reached their mining employment peaks at different times.

Individual County Trajectories

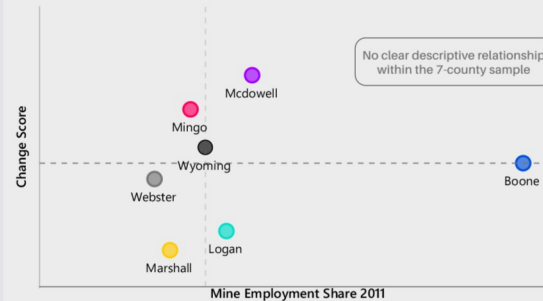
County	● Own Peak Year	Peak Mining Emp.	Latest Mining Emp.	% Decline
Logan	2011	2128	1879	-12%
Marshall	2014	1722	1279	-26%
Wyoming	2013	1301	1261	-3%
Mcdowell	2011	1209	777	-36%
Boone	2008	3913	623	-84%
Mingo	2002	1683	358	-79%
Webster	2002	488	40	-92%

Source: MSHA Mine Employment Data. % decline is calculated from each county's individual mining employment peak year to the latest available year in the dataset.

Did Counties Change Similarly After 2011?

The seven counties had substantial mining employment shares in 2011 but experienced different levels of change afterward.

Mining Employment Share (2011) vs. Change Score



Reference lines indicate sample medians.

How Did Selected Indicators Change?

Labor-market and demographic indicators varied across the seven counties.

Labor-Market and Demographic Change, 2011–2024

County	Change Score	Labor Force Exit	Emp-To-Pop Decline	Pop Decline	Demographic Aging
Mcdowell	86 (High)	23.54	10.40	19.4%	5.45
Mingo	78 (High)	15.80	3.02	16.1%	8.37
Wyoming	70 (Moderate)	12.54	0.48	13.4%	8.55
Webster	63 (Moderate)	11.78	0.69	11.5%	7.51
Boone	67 (Moderate)	10.77	0.16	15.0%	8.04
Marshall	48 (Low)	6.50	-1.61	9.9%	6.36
Logan	52 (Low)	5.72	-3.90	14.1%	7.43

Change Score | High Moderate Low

Change Score is an exploratory composite (0–100) combining selected labor-market and demographic indicators.



Key Takeaway

All seven counties had substantial mining dependence in 2011, but they experienced different patterns of mining employment decline and subsequent labor-market and demographic change.



Policy Implication

The differences across counties suggest that policies addressing coal-dependent communities should consider local labor-market and demographic conditions alongside mining dependence instead of relying on mining employment alone.